

wooclap

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Lecture 08A

Before-Tutorial-Session

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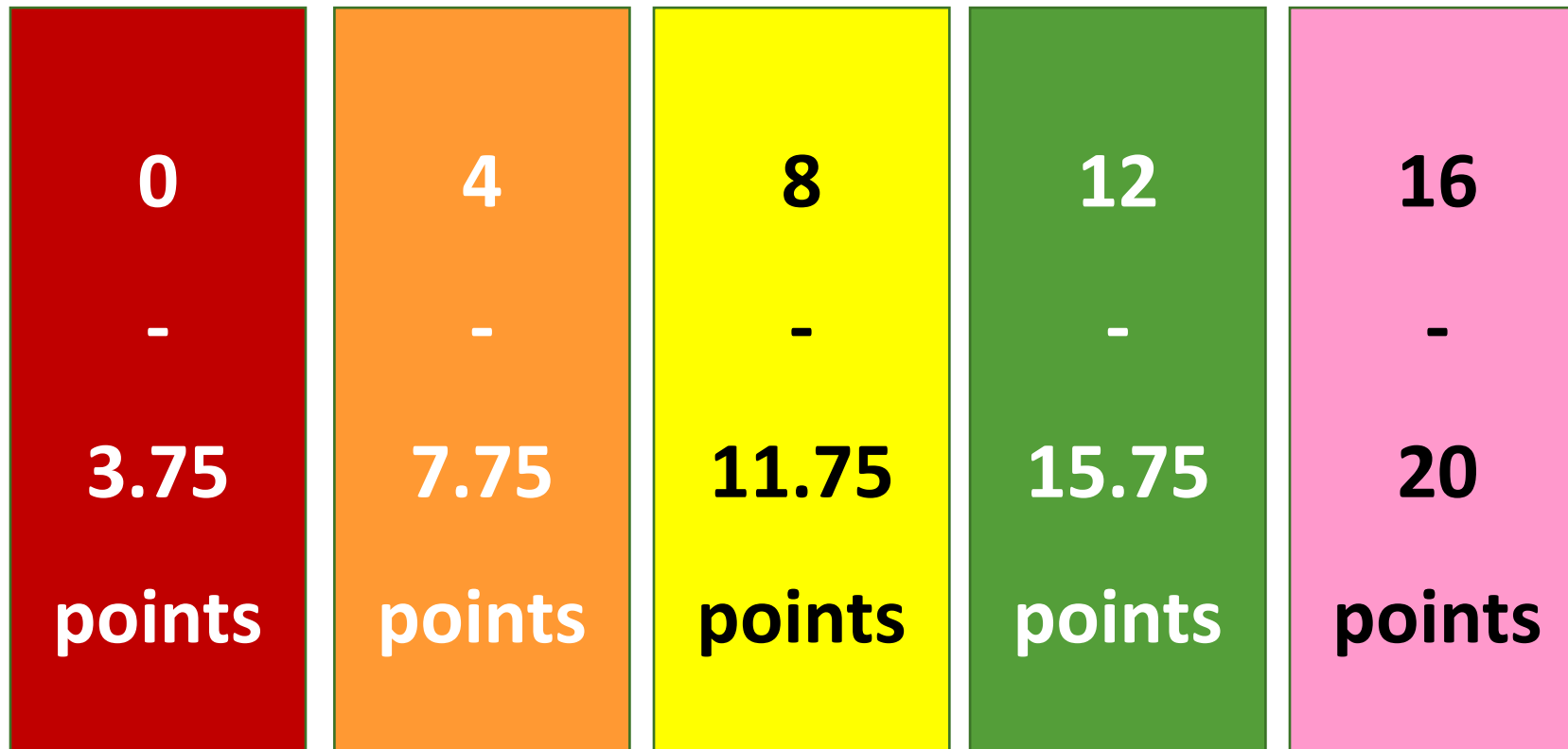
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The Wooclap scores are available on NTULearn.

Which box do you belong to?

To obtain one point, you need to answer TRUTHFULLY.



Let's do this question. Tut06, Q8

Question 8. Consider the following matrix $\mathbf{A}$ and basis $\mathcal{A}$ for $\mathbb{R}^3$, where

$$\mathbf{A} = \begin{bmatrix} 13 & 1 & 4 \\ 1 & 13 & 4 \\ 4 & 4 & 10 \end{bmatrix} \text{ and } \mathcal{A} = \left\{ \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

- (a) Find the change-of-basis matrix from $\mathcal{A}$ to the standard basis $\mathcal{B}$ for $\mathbb{R}^3$.
- (b) Let $\mathbf{T}$ be the matrix-multiplication corresponding to $\mathbf{A}$. Determine $[\mathbf{T}]_{\mathcal{A}}^{\mathcal{A}}$.

Question 8. Consider the following matrix A and basis $\mathcal{A}$ for $\mathbb{R}^3$, where

$$\hat{T}(v) = \hat{A}v \quad A = \begin{bmatrix} 13 & 1 & 4 \\ 1 & 13 & 4 \\ 4 & 4 & 10 \end{bmatrix} \text{ and } \mathcal{A} = \left\{ \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

(a) Find the change-of-basis matrix from $\mathcal{A}$ to the standard basis $\mathcal{B}$ for $\mathbb{R}^3$.

How are the columns and rows of the change-of-basis matrix indexed?

$$\begin{bmatrix} \hat{1} \\ \hat{1} \end{bmatrix}_{\mathcal{A}}^{\mathcal{B}} = \left[\underbrace{\quad}_{\text{"domain"} \rightarrow \mathcal{A} \text{ or } \mathcal{B}} \right]_{\mathcal{A} \text{ or } \mathcal{B}}^{\text{"codomain"} \rightarrow \mathcal{B}}$$

v written in basis $\mathcal{A}$
 $\rightarrow$ written in basis $\mathcal{B}$.

columns are indexed by $\mathcal{A}$ and rows are indexed by $\mathcal{B}$.

Question 8. Consider the following matrix A and basis $\mathcal{A}$ for $\mathbb{R}^3$, where

$$A = \begin{bmatrix} 13 & 1 & 4 \\ 1 & 13 & 4 \\ 4 & 4 & 10 \end{bmatrix} \text{ and } \mathcal{A} = \left\{ \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

(a) Find the change-of-basis matrix from $\mathcal{A}$ to the standard basis $\mathcal{B}$ for $\mathbb{R}^3$.

$$[\hat{I}]_{\mathcal{A}}^{\mathcal{B}} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ -2 & 0 & 1 \end{bmatrix} \begin{matrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \end{matrix}$$

$$\begin{matrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \end{matrix}$$

$$I \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$$

$$\vdots$$

Question 8. Consider the following matrix A and basis $\mathcal{A}$ for $\mathbb{R}^3$, where

$$A = \begin{bmatrix} 13 & 1 & 4 \\ 1 & 13 & 4 \\ 4 & 4 & 10 \end{bmatrix} \text{ and } \mathcal{A} = \left\{ \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

(b) Let T be the matrix-multiplication corresponding to A . Determine $[T]_{\mathcal{A}}^{\mathcal{A}}$.

Which is the correct formula for $[T]_{\mathcal{A}}^{\mathcal{A}}$?

✓ 1. $[T]_{\mathcal{A}}^{\mathcal{A}} = [I]_{\mathcal{B}}^{\mathcal{A}} [T]_{\mathcal{B}}^{\mathcal{B}} [I]_{\mathcal{A}}^{\mathcal{B}}$

$\begin{bmatrix} \uparrow \\ \uparrow \end{bmatrix}_{\mathcal{A}}^{\mathcal{A}}$ ← end
← start

2. $[T]_{\mathcal{A}}^{\mathcal{A}} = [I]_{\mathcal{A}}^{\mathcal{A}} [T]_{\mathcal{B}}^{\mathcal{B}} [I]_{\mathcal{A}}^{\mathcal{B}}$

3. $[T]_{\mathcal{A}}^{\mathcal{A}} = [I]_{\mathcal{B}}^{\mathcal{A}} [T]_{\mathcal{B}}^{\mathcal{B}} [I]_{\mathcal{B}}^{\mathcal{A}}$ → never start from

4. $[T]_{\mathcal{A}}^{\mathcal{A}} = [I]_{\mathcal{A}}^{\mathcal{B}} [T]_{\mathcal{B}}^{\mathcal{B}} [I]_{\mathcal{B}}^{\mathcal{A}}$ → A

→ options 3 and 4 are wrong because should start from $\mathcal{A}$.

Question 8. Consider the following matrix A and basis $\mathcal{A}$ for $\mathbb{R}^3$, where

$$A = \begin{bmatrix} 13 & 1 & 4 \\ 1 & 13 & 4 \\ 4 & 4 & 10 \end{bmatrix} \text{ and } \mathcal{A} = \left\{ \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

(b) Let T be the matrix-multiplication corresponding to A . Determine $[T]_{\mathcal{A}}^{\mathcal{A}}$.

Tutorial Solution: We use the formula:

$$[T]_{\mathcal{A}}^{\mathcal{A}} = [I]_{\mathcal{B}}^{\mathcal{A}} [T]_{\mathcal{B}}^{\mathcal{B}} [I]_{\mathcal{A}}^{\mathcal{B}}$$

with

$$[I]_{\mathcal{A}}^{\mathcal{B}} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ -2 & 0 & 1 \end{bmatrix},$$

(a)

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{bmatrix} 13 & 1 & 4 \\ 1 & 13 & 4 \\ 4 & 4 & 10 \end{bmatrix},$$

why? ask your tutors

$$[I]_{\mathcal{B}}^{\mathcal{A}} = ([I]_{\mathcal{A}}^{\mathcal{B}})^{-1}$$

- we compute

$[T]_{\mathcal{A}}^{\mathcal{A}}$ another way.

- objective:

→ motivate "diagonalisation"

→ revise "matrix multiplication"

Question 8. Consider the following matrix A and basis $\mathcal{A}$ for $\mathbb{R}^3$, where

$$A = \begin{bmatrix} 13 & 1 & 4 \\ 1 & 13 & 4 \\ 4 & 4 & 10 \end{bmatrix} \text{ and } \mathcal{A} = \left\{ \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

(b) Let T be the matrix-multiplication corresponding to A . Determine $[T]_{\mathcal{A}}^{\mathcal{A}}$.

Again, we use the formula:

$$\underbrace{[T]_{\mathcal{A}}^{\mathcal{A}}}_{\hat{D}} = \underbrace{[I]_{\mathcal{B}}^{\mathcal{A}}}_{\hat{P}^{-1}} \underbrace{[T]_{\mathcal{B}}^{\mathcal{B}}}_{\hat{A}} \underbrace{[I]_{\mathcal{A}}^{\mathcal{B}}}_{\hat{P}}$$

with

$$[I]_{\mathcal{A}}^{\mathcal{B}} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ -2 & 0 & 1 \end{bmatrix},$$

$\hat{P}$.

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{bmatrix} 13 & 1 & 4 \\ 1 & 13 & 4 \\ 4 & 4 & 10 \end{bmatrix},$$

$$[I]_{\mathcal{B}}^{\mathcal{A}} = ([I]_{\mathcal{A}}^{\mathcal{B}})^{-1}$$

$\hat{P}^{-1}$

$$\text{So } \hat{D} = \hat{P}^{-1} \hat{A} \hat{P}$$

Revision. Matrix Multiplication.

Which of the following is equivalent to $D = P^{-1}AP$?

1. $AP = DP$

2. $AP = PD$

$$\hat{P} \hat{D} = \hat{P}^{-1} \hat{A} \hat{P}$$

! order does matter

$$\hat{A}\hat{B} \neq \hat{B}\hat{A}$$

Revision. Matrix Multiplication.

c_i 's are columns

r_i 's are rows

$$\text{Suppose } P = \begin{bmatrix} \uparrow & \uparrow & \uparrow \\ c_1 & c_2 & c_3 \\ \downarrow & \downarrow & \downarrow \end{bmatrix} = \begin{bmatrix} \leftarrow & r_1 & \rightarrow \\ \leftarrow & r_2 & \rightarrow \\ \leftarrow & r_3 & \rightarrow \end{bmatrix}.$$

Which of the following is AP ?

✓ 1. $AP = \begin{bmatrix} \uparrow & \uparrow & \uparrow \\ Ac_1 & Ac_2 & Ac_3 \\ \downarrow & \downarrow & \downarrow \end{bmatrix}$ $\hat{A}$ c_i $\rightarrow$ ok!
 3×3 3×1

2. $AP = \begin{bmatrix} \leftarrow & Ar_1 & \rightarrow \\ \leftarrow & Ar_2 & \rightarrow \\ \leftarrow & Ar_3 & \rightarrow \end{bmatrix}$ $\hat{A}$ r_i $\rightarrow$ wrong!
 3×3 1×3

Revision. Matrix Multiplication.

Let $A = \begin{bmatrix} \uparrow & \uparrow & \uparrow \\ \mathbf{y}_1 & \mathbf{y}_2 & \mathbf{y}_3 \\ \downarrow & \downarrow & \downarrow \end{bmatrix} = \begin{bmatrix} \leftarrow & \mathbf{x}_1 & \rightarrow \\ \leftarrow & \mathbf{x}_2 & \rightarrow \\ \leftarrow & \mathbf{x}_3 & \rightarrow \end{bmatrix}$ and $\mathbf{c} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{bmatrix}$. $\hat{A} \hat{c}$

$y_i = 3 \times 1$ $x_i = 1 \times 3$

Which of the following is $A\mathbf{c}$?

1. $A\mathbf{c} = \mu_1 \mathbf{y}_1 + \mu_2 \mathbf{y}_2 + \mu_3 \mathbf{y}_3 \rightarrow$ linear combi of cols

2. $A\mathbf{c} = \mu_1 \mathbf{x}_1 + \mu_2 \mathbf{x}_2 + \mu_3 \mathbf{x}_3 \rightarrow$ linear combi of rows

$\hat{A} \hat{c} \rightarrow \hat{A} \hat{c} \Rightarrow 3 \times 1$

$\downarrow \quad \downarrow$
 $3 \times 3 \quad 3 \times 1$

Revision. Matrix Multiplication.

Let $\mathbf{P} = \begin{bmatrix} \uparrow & \uparrow & \uparrow \\ \mathbf{c}_1 & \mathbf{c}_2 & \mathbf{c}_3 \\ \downarrow & \downarrow & \downarrow \end{bmatrix} = \begin{bmatrix} \leftarrow & \mathbf{r}_1 & \rightarrow \\ \leftarrow & \mathbf{r}_2 & \rightarrow \\ \leftarrow & \mathbf{r}_3 & \rightarrow \end{bmatrix}$ and $\mathbf{D} = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$.

Which of the following is $\mathbf{PD}$?

✓ 1. $\mathbf{PD} = \begin{bmatrix} \uparrow & \uparrow & \uparrow \\ \lambda_1 \mathbf{c}_1 & \lambda_2 \mathbf{c}_2 & \lambda_3 \mathbf{c}_3 \\ \downarrow & \downarrow & \downarrow \end{bmatrix}$

2. $\mathbf{PD} = \begin{bmatrix} \leftarrow & \lambda_1 \mathbf{r}_1 & \rightarrow \\ \leftarrow & \lambda_2 \mathbf{r}_2 & \rightarrow \\ \leftarrow & \lambda_3 \mathbf{r}_3 & \rightarrow \end{bmatrix}$

$$\begin{aligned} \hat{\mathbf{P}} \hat{\mathbf{D}} &= \hat{\mathbf{P}} \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \\ &= \left[\hat{\mathbf{P}} \begin{bmatrix} \lambda_1 \\ 0 \\ 0 \end{bmatrix} \quad \hat{\mathbf{P}} \begin{bmatrix} 0 \\ \lambda_2 \\ 0 \end{bmatrix} \quad \hat{\mathbf{P}} \begin{bmatrix} 0 \\ 0 \\ \lambda_3 \end{bmatrix} \right] \\ &= \begin{bmatrix} \uparrow & \uparrow & \uparrow \\ \lambda_1 \mathbf{c}_1 & \lambda_2 \mathbf{c}_2 & \lambda_3 \mathbf{c}_3 \\ \downarrow & \downarrow & \downarrow \end{bmatrix} \end{aligned}$$

Revision. Matrix Multiplication.

Since $A = \begin{bmatrix} 13 & 1 & 4 \\ 1 & 13 & 4 \\ 4 & 4 & 10 \end{bmatrix}$ and $P = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ -2 & 0 & 1 \end{bmatrix}$,

we have ...

$$\hat{A}\hat{P} = \begin{bmatrix} 6 & 12 & 18 \\ 6 & -12 & 18 \\ -12 & 0 & 18 \end{bmatrix} = \begin{bmatrix} 6 \cdot 1 & 12 \cdot 1 & 18 \cdot 1 \\ 6 \cdot 1 & 12 \cdot (-1) & 18 \cdot 1 \\ 6 \cdot (-2) & 12 \cdot 0 & 18 \cdot 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 6 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 18 \end{bmatrix}$$

$$= \hat{P} \hat{D}$$

Recall: we wanted to find

$$\hat{D} = \hat{P}^{-1} \hat{A} \hat{P}$$

$$\hat{A} \hat{P} = \hat{P} \hat{D}$$

col 1

$$\begin{bmatrix} 13+1-8 \\ 1+13-8 \\ 4+4-20 \end{bmatrix}$$

$\hat{D}$ appears nice because

$$\left. \begin{aligned} \hat{A} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} &= 6 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \\ \hat{A} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} &= 12 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \\ \hat{A} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} &= 18 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \end{aligned} \right\} \begin{array}{l} \text{special name:} \\ - \text{eigenvector} \\ - \text{eigenvalue} \end{array}$$

Eigenvalues and Eigenvector

Definition. Let A be an $n \times n$ -matrix.

A nonzero $\underline{v} \in \mathbb{R}^n$ is an eigenvector of A if

$$Av = \lambda v \text{ for some } \lambda \in \mathbb{R}.$$

Here, we say λ is the eigenvalue corresponding to eigenvector v .

↙
aka eigenvalue - eigenvector pair

* Go through in detail on
Thursday

Example. Consider $A = \begin{bmatrix} 13 & 1 & 4 \\ 1 & 13 & 4 \\ 4 & 4 & 10 \end{bmatrix}$.

Show that $\begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$, $\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ are eigenvectors of A .

For each eigenvector, determine the corresponding eigenvalues.

Definition.

Let A be an $n \times n$ -matrix.

A nonzero $v \in \mathbb{R}^n$ is an eigenvector of A if

$$Av = \lambda v \text{ for some } \lambda \in \mathbb{R}.$$

Here, we say λ is the eigenvalue corresponding to eigenvector v .

Solution: This is because

$$A \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} = 6 \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}, \quad A \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = 12 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 18 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

corr. eigenvalue 6

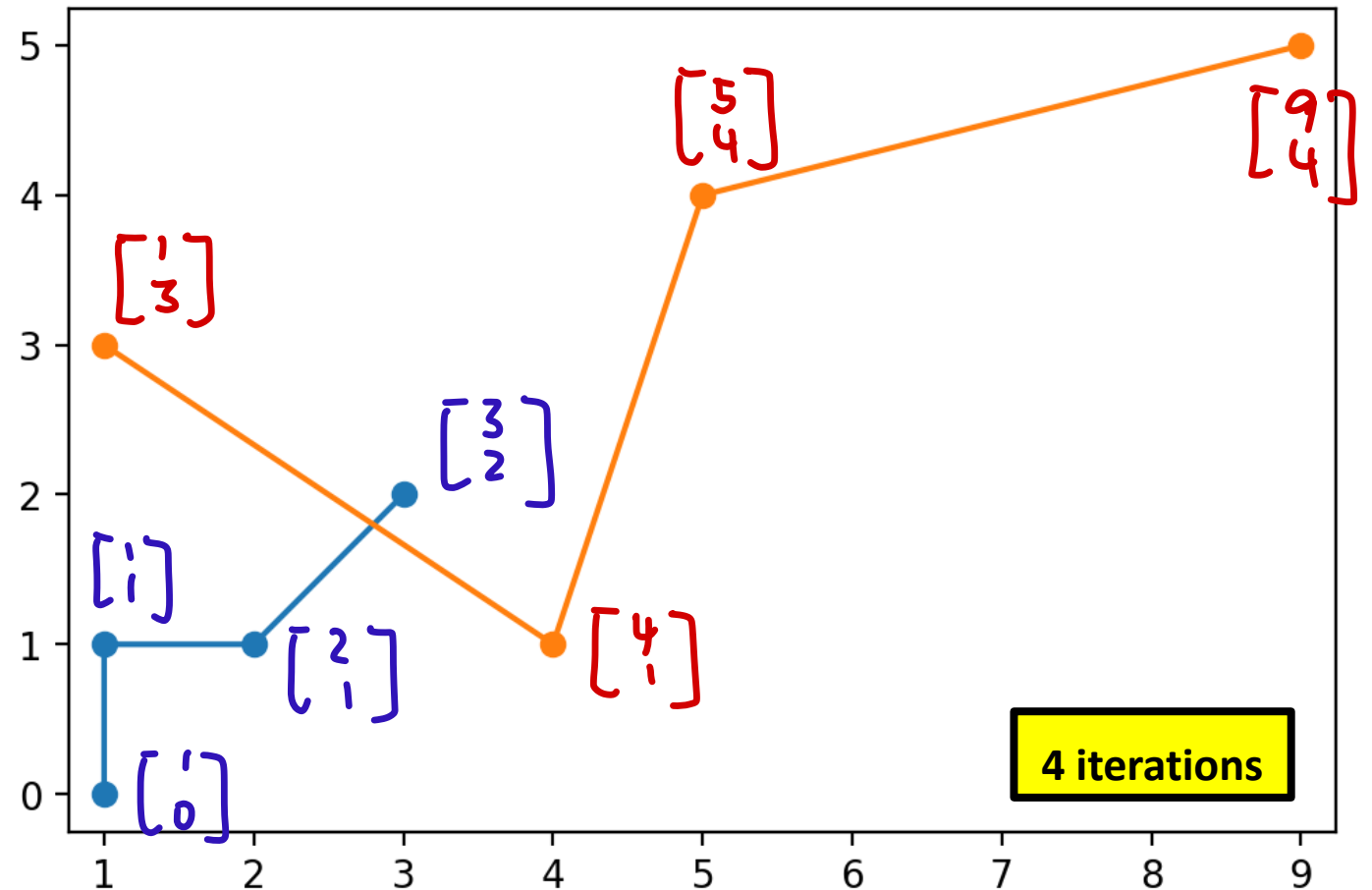
corr. eigenvalue 12

corr. eigenvalue 18

Why Eigenvectors? Power Iteration

- Let $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$.
- Start with any point $\begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$.
- Let us compute

$$\begin{bmatrix} x_0 \\ y_0 \end{bmatrix}, A \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}, A^2 \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}, \dots$$

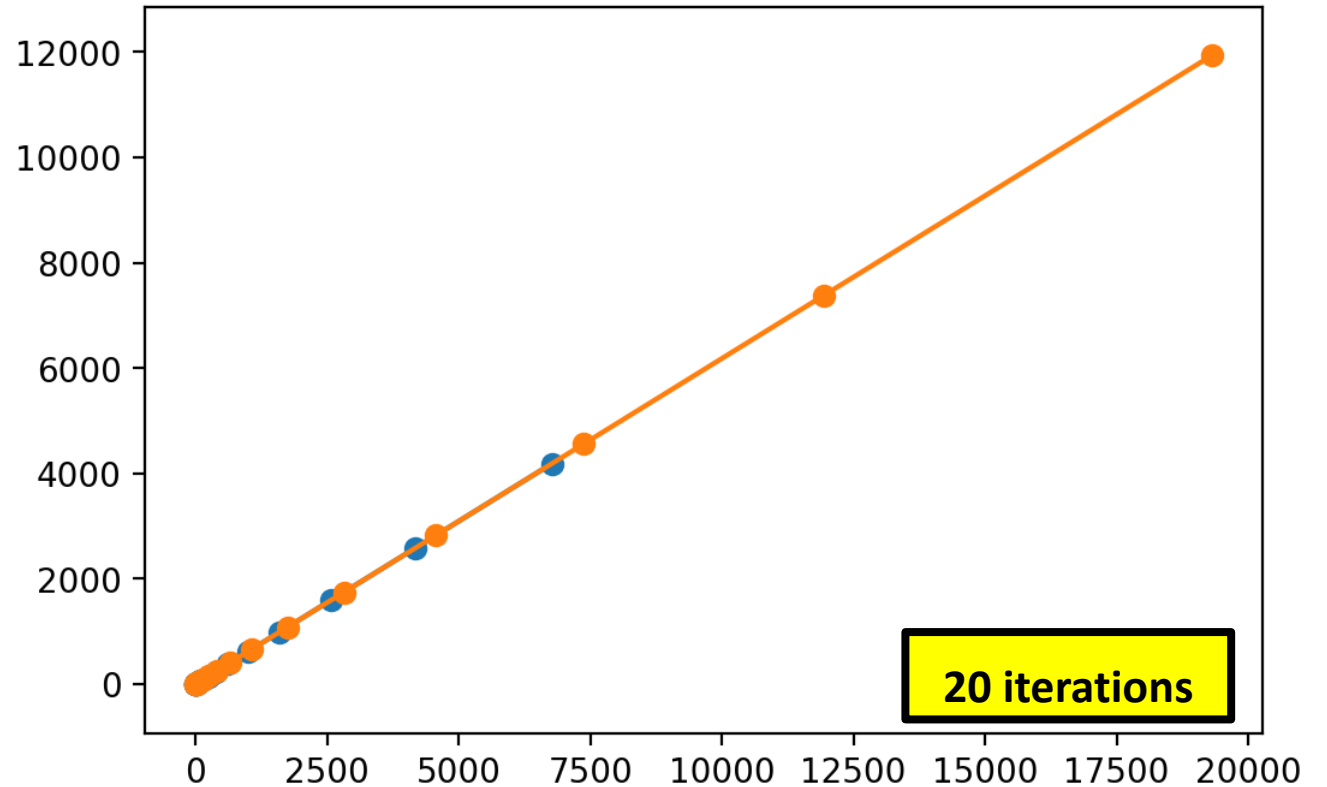
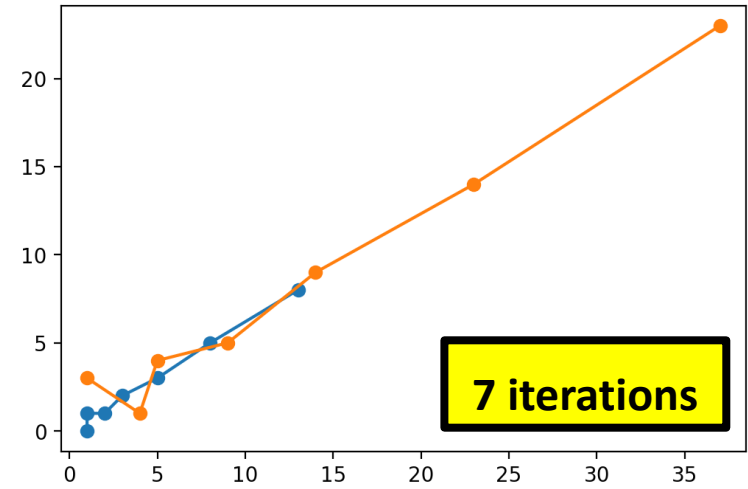
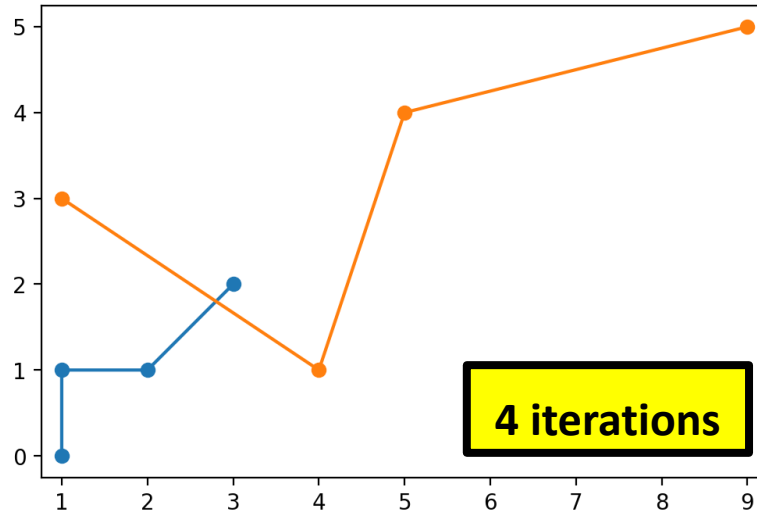


Power Iteration

- Let $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$.
- Start with any point $\begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$.
- Let us compute

$$\begin{bmatrix} x_0 \\ y_0 \end{bmatrix}, A \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}, A^2 \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}, \dots$$

Both trails follow the
same direction



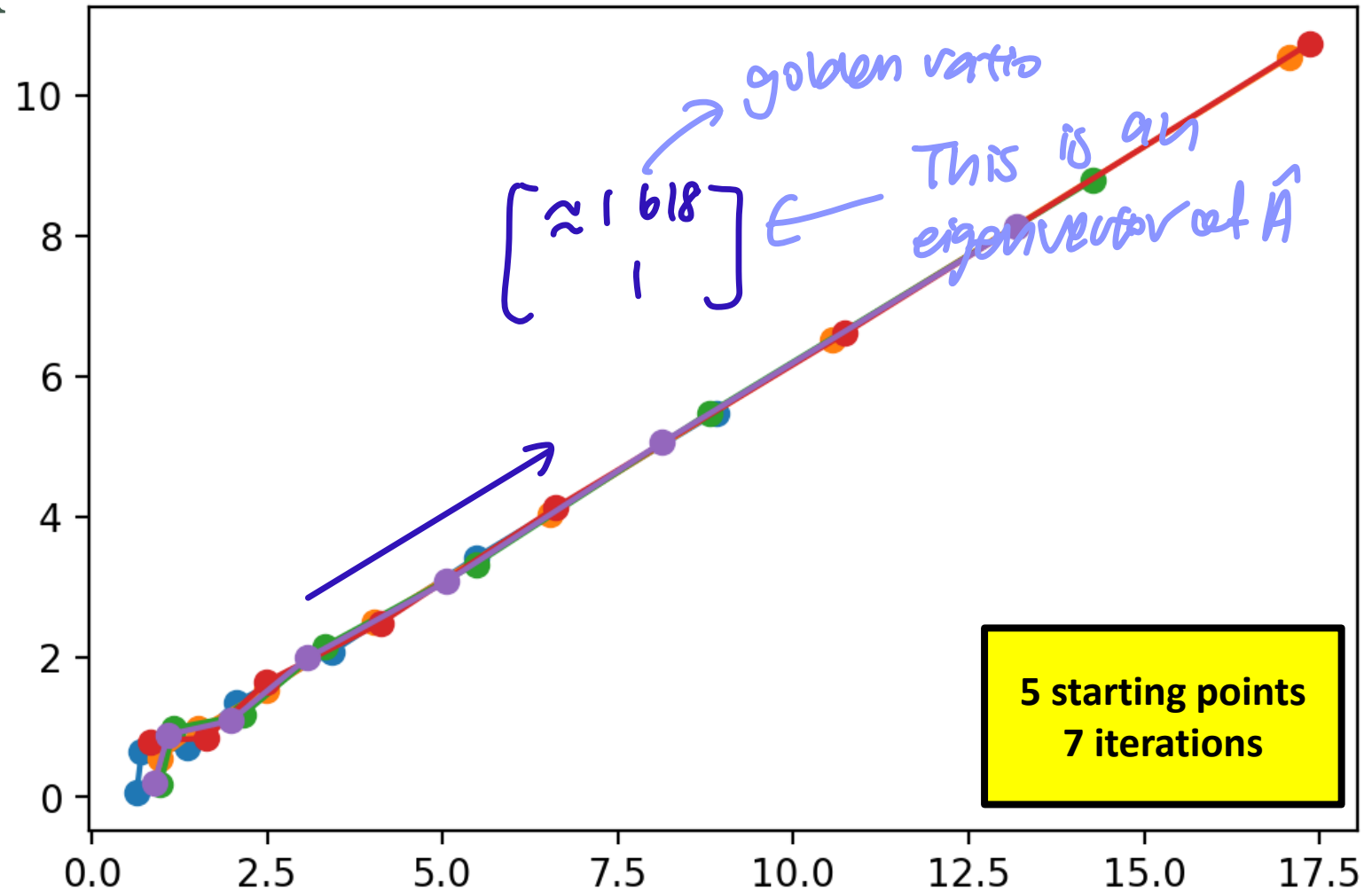
Power Iteration

- Let $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$.
- Start with any point $\begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$.
- Let us compute

$$\begin{bmatrix} x_0 \\ y_0 \end{bmatrix}, A \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}, A^2 \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}, \dots$$

Next lesson:

how to find eigenvectors



Are you going to tutorial?

Yes, you are!

You will be collecting your midterm script.